

Referee Report - Towards identifying possible fault-tolerant advantage of quantum linear system algorithms in terms of space, time and energy

March 10, 2026

The authors provide a detailed resource estimation for the HHL quantum linear system algorithm running on a fault-tolerant superconducting device, including energy cost estimates. Logical resources (T-gate counts, qubit counts) are estimated via a Q# implementation, and Trotter and QFT errors are bounded using exponential-type error bounds from Ref. [13]. These logical resources are then mapped to physical resources using a surface code optimization framework, and the authors compare the resulting quantum overhead against the classical conjugate gradient (CG) method, reporting possible quantum advantages at matrix sizes $N \sim 2^{33}$ to 2^{48} , requiring $O(10^5)$ physical qubits. Energy estimates are based on IBM Quantum System Two.

The central question in this work, namely the energy and quantum resource cost crossover point for an important quantum algorithm, is an important and timely question for the community. Overall, the authors provide a useful methodology for performing detailed complexity estimates across time, space, and energy. The error analysis using exponential-type error bounds is interesting and valuable to the community. However, in its current state, I do not believe the paper is suitable for publication without major revisions or clarifications. Several assumptions underlying the crossover claims with classical methods need to be clarified and discrepancies between the main text and appendices need to be addressed for the results to be reliable.

Major Comments

1. The use of the original HHL algorithm as opposed to state-of-the-art (SOTA) QLSA reduces the practical relevance of the results. The authors acknowledge that “polynomial scalings on s , κ , $1/\varepsilon$ has been updated for the convenience of exact pre-factor derivation” (p. 3), implying that they intentionally use the original HHL framework with $O(\kappa^2 s/\varepsilon^2)$ scaling rather than modern QLSA variants (Refs. [33–37]) that achieve dramatically improved dependence on the condition number (e.g., linear in κ). However, since the central result of the work is a crossover analysis between quantum and classical methods, using an outdated algorithm reduces the relevance of the claimed advantage boundaries. At a minimum, the authors should discuss/estimate the shift in the crossover points with SOTA algorithms.
2. The authors exclude data loading and readout costs from the resource estimates. However, efficient data loading remains a major challenge for general classes of matrices, and hence the claimed advantage boundary could shift significantly when these costs are included. The caption for Figure 1 should caution the readers regarding this assumption. Additionally, it would

strengthen the paper to provide even rough estimates of how the crossover points would shift under realistic data loading models.

3. The comparison with classical baselines is not completely clear. The main text refers to the conjugate gradient (CG) method as the equivalent classical baseline, which requires symmetric positive definite (SPD) systems, as noted in Appendix D. However, the analysis in Appendix D is based on CGNE. These are different algorithms, as CGNE squares the condition number and requires more flops per iteration. It appears that this square factor was included in Thm. 4, but this is not readily apparent from the text. Moreover, the analysis seems to mix different classes of matrices – some positive semi-definite, some symmetric. The authors should clarify whether or not they are assuming the input matrix is Hermitian during the classical baseline. If so, along with the assumption on the eigenvalues of A , CG is a reasonable comparison, but in the appendix the authors compare with CGNE. Regardless, one should ensure a comparison against state-of-the-art classical methods including preconditioned, sparse solvers, rather than costly normal equation-based methods. All in all, the comparison with classical methods is not clear and does not seem to be properly handled between the body and the appendices.
4. The classical CG runtime is estimated on a single desktop CPU at 1 GHz / 50 W, while the quantum side uses an optimized fault-tolerant architecture with $O(10^5)$ physical qubits. This is not an apples-to-apples comparison. Appendix D compares against Cholesky decomposition on supercomputers, pushing the crossover to $\sim 2^{100}$, yet this result appears only in the appendix. The Cholesky/supercomputer comparison should be featured in the main text and the main text should clearly state that the advantage boundaries would depend on the classical hardware used. Moreover, the classical hardware assumed for comparison is likely not representative of the hardware that would likely be used for these classes of problems. The energy cost for quantum hardware is estimated using numbers found in Ref. 18, but these numbers are based on *current* estimates and are, in the words of the authors of Ref. 18, “*extremely* rough”. These numbers also ignore economies of scale that are likely afforded by larger system sizes.
5. The probabilistic bound (Eq. B19) used to prove Statement 1 is derived from numerical experiments on “small matrix size” (pg. 17), with the claim that “the error does not depend on n .” This claim is crucial for the entire scaling analysis, yet: (a) Figure 9 lacks axis labels and a description of the experimental setup; (b) the matrix sizes, distributions, and number of samples are not reported; (c) no rigorous justification is given for why small-matrix behavior should extrapolate to larger matrices. This needs to be explained clearly.
6. Figure 1 assumes $\kappa = s = \log(N)$. (1) This is an optimistic assumption on κ that may only be valid in limited settings. Many real-world linear systems have condition numbers that grow polynomially with N . Since the quantum cost scales as κ^2 and CG scales linearly in κ , the crossover point is very sensitive to this assumption. The authors should validate this assumption using examples of matrix classes where this holds true. It would also strengthen the paper to present crossover estimates under more realistic assumptions (e.g., $\kappa = N^\alpha$). (2) The assumption on $s = \log(N)$ is quite interesting. The sparsity is usually a fixed property of the matrix structure and not a function of its size. The authors should again validate this assumption using examples of real matrix classes.

Minor Comments

1. Many of the relevant symbols are either ill defined, overloaded, or not defined. For example, T represents at least four distinct quantities throughout: the number of T -gates, the SWAP

operator, the number of repeated unitaries, and the total operating time.

2. In pg. 17, Fig. 19, the axes are unlabeled and the caption does not describe the experimental setup (matrix types, sizes, number of samples).
3. Multiple grammatical and spelling errors are found throughout the paper. A few examples: “primarily” (p. 3), “physicla” (p. 5, p. 22), “Toffli” (throughout; should be “Toffoli”), “tyler expansion” (p. 15; should be “Taylor”), “dividingng” (Appendix D), “changable” (p. 3), “Tranvis” (p. 6). Please check for typos and grammatical errors.
4. The limitations and assumptions used in the work such as exclusion of data loading and oracle costs, use of original HHL, and classical baseline on 1 CPU, should be restated in the Conclusions.